

Coding & Solution

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Team ID	SWTID-2026-9792
Project Name	FitBuddy – AI Fitness Plan Generator

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Core implementation flow

1. User submits profile data → 2. FastAPI checks User ID → 3. Gemini generates a 7-day workout plan → 4. Gemini generates a nutrition/recovery tip → 5. User and plan are stored in SQLite → 6. Result page displays the generated plan → 7. Feedback can trigger a revised plan → 8. Admin can view stored records after authentication.

Representative implementation modules

Module	Implemented Logic
Workout generation	<code>generate_workout_gemini(username, age, weight, goal, intensity)</code>
Nutrition generation	<code>generate_nutrition_tip_with_flash(goal)</code>
Plan revision	<code>update_workout_plan(original_plan, feedback, username, goal, intensity)</code>
Persistence	User and Plan SQLAlchemy models with save/get/update helpers.
Admin	Environment credentials + session flag + protected <code>/view-all-users</code> route.
Health	<code>/health</code> endpoint returns application status.

Note: API keys and credentials are not embedded in the documentation or source excerpts; they are expected through environment configuration.